Updating the Raspberry Pi with New Code

This guide explains how to update the Raspberry Pi with the latest version of the TestStandWebInterface code from your personal PC. Follow the steps below to ensure a smooth update.

Prerequisites

- Network Connection: Ensure your Raspberry Pi is connected to the network.
- WinSCP Installed: Install and configure WinSCP on your PC.
- Thonny Available: Confirm that Thonny (Python IDE) is installed on the Raspberry Pi.
- Administrative Access: You have the necessary permissions on the Raspberry Pi to modify files and update the crontab.

Step 1: Create a New Folder on the Raspberry Pi

1. Folder Creation:
   a. Create a new folder on the Raspberry Pi using the naming format:
      i.
      ```
      YYYY-MM-DD_WebInterface
      ```

Step 2: Transfer the Code Using WinSCP

2. Connect via WinSCP:
   a. Open WinSCP on your PC and connect to the Raspberry Pi.
3. Copy the Code Folder:
   a. Copy the TestStandWebInterface folder from your PC.
4. Paste into New Folder:
   a. Paste the TestStandWebInterface folder into the new folder you created on the Raspberry Pi.

Step 3: Update the app.py File

5. Open app.py:
   a. Navigate to the new folder, locate app.py, right-click on it, and choose “Open in Thonny.”
6. Comment Out Unused Imports:
   a. In app.py, comment out the lines:
      i.
      ```
      import wexpect
      from wexpect import ExceptionPexpect
      ```
7. Update Environment Variables and Paths:
   a. Change:

```
base_path = os.getenv('csv_path')
```

b. To:

```
base_path = os.getenv('LOCAL_BASE_PATH')
```

c. Add:

```
network_base_path = os.getenv('NETWORK_BASE_PATH')
```

8. Set Additional Paths:
   a. Add the following line to define the Excel directory

```
xlsx_dir = os.path.join(network_base_path, 'can_data_local.db')
```

   b. Add another line for the database backups:

```
db_backup_dir = os.path.join(network_base_path, 'db_backups')
```

9. Modify the Process Target:
   a. Change:

```
(target=process_can_data)
```

   b. To:

```
(target=process_can_data_locally)
```

10. Configure the Server Host:
    a. Change:

```
app.run(debug=debug_mode, port=port)
```

    b. To:

```
app.run(debug=debug_mode, host='192.168.1.90', port=port)
```

Step 4: Update the databaseManipulation.py File

11. Open databaseManipulation.py
    a. Open this file in Thonny.
12. Update Environment Variable:
    a. Change:

```
base_path = os.getenv('csv_path')
```

    b. To:

```
base_path = os.getenv('LOCAL_BASE_PATH')
```

13. Add Network Base Path:

a. Add the following line:

```
network_base_path = os.getenv('NETWORK_BASE_PATH')
```

Step 5: Update the decodeAndProcess.py File

14. Open decodeAndProcess.py:
    a. Open this file in Thonny.
15. Comment Out wexpect Import:
    a. Comment out:

```
import wexpect
```

Step 6: Configure the Environment Variables (.env File)

16. Locate the .env File:
    a. Using the Raspberry Pi file explorer, navigate to:

```
/home/devteam/Pyhon/<new_folder>
```

17. Find the .env file, right-click it, and select “Open with Text Editor.”
18. Update the .env Content:
    a. Ensure the file contains exactly the following:

```
LOCAL_BASE_PATH=/home/devteam/Python/<new_folder>/TestStandWebInterface
NETWORK_BASE_PATH=/home/devteam/sambashare
```

    i. *Tip:* If you are unsure of the local path, copy it directly from the file explorer’s address bar.
19. Save Changes:
    a. Once verified, save the changes in your text editor.

Step 7: Update the Crontab Entry or Systemd script

20. Open Terminal:
    a. Open a new terminal window on the Raspberry Pi.
21. Edit Crontab:
    a. Run:

```
crontab -e
```

22. Modify the @reboot Path:
    a. Scroll to the bottom of the file using the arrow keys.
    b. Locate the line beginning with @reboot and update the file (starting with /home/devteam/Python/) to reflect the path to your new folder.

23. Save and Exit:
    a. Press Ctrl + X to exit
    b. Press Y To confirm saving, then hit Enter
24. Alternatively, we can use Systemd instead of crontab. This will make it easier to read debug messages from the logs, and make it easier to restart the service.
25. Edit (or create) a systemd script:
    a. sudo nano /etc/systemd/system/teststand-web.service

```
[Unit]
Description=TestStand Web Interface
After=network.target
Wants=network-online.target

[Service]
Type=simple
User=devteam
WorkingDirectory=/home/devteam/Python/2025-08-25_WebInterface/TestStandWebInterface
ExecStart=/usr/bin/python3 /home/devteam/Python/2025-08-25_WebInterface/TestStandWebInterface/app.py
Restart=always
RestartSec=5
StandardOutput=journal
StandardError=journal
Environment=PYTHONUNBUFFERED=1

[Install]
WantedBy=multi-user.target

```

    b. use ctrl-k to copy the line with WorkingDirectory and use ctrl-u twice to paste it twice to duplicate the line.
    c. do the same for the line with ExecStart.
    d. use the # character to comment out the older versions of those lines, and edit the new versions to show the new file locations.
26. Enable and restart the teststand systemd service
    a.
    ```
    sudo systemctl enable teststand-web.service
    ```
    b.
    ```
    sudo systemctl enable teststand-web.service
    ```
    c.
    ```
    sudo systemctl enable teststand-web.service
    ```

Step 8: Reboot and Verify the Update

27. Reboot the Raspberry Pi:
    a. Run the command:

```
sudo reboot
```

28. Test the Update:
    a. After the Pi restarts, open a web browser and navigate to:

```
http://raspberrypi:8000/
```

b. If the website loads successfully, your update was successful.

29. To check the log for troubleshooting purposes:

    a.
    ```
    journalctl -u teststand-web.servoice -n 50 --no-pager
    ```